

Infectious Diseases

Introduction

- Infectious diseases are caused by pathogens (bacteria, viruses, fungi, etc.) that disrupt the body's balance (homeostasis).
- This lesson covers how diseases spread, how they are identified, and how they are treated.

1. Pathogens & Germ Theory

- Pathogens are microorganisms that cause disease.
- Koch's Postulates provide a method to prove a pathogen causes a specific disease:
 - a. Found in sick individuals.
 - b. Grown in lab culture.
 - c. Causes disease in healthy host.
 - d. Re-isolated and matches original.

2. Disease Reservoirs & Transmission

- **Reservoirs:** Places where pathogens live (humans, animals, soil, water).
- **Transmission:** Can happen through:
 - Direct contact
 - Airborne droplets
 - Contaminated surfaces
 - Vectors (e.g., mosquitoes)

3. Disease Patterns

- **Endemic:** Constantly present in a population.
- **Epidemic:** Sudden outbreak in one area.
- **Pandemic:** Widespread global outbreak.

4. Antibiotics & Resistance

- Antibiotics kill or inhibit bacterial growth (e.g., penicillin).
- Antibiotic resistance is rising, making treatment harder for diseases like MRSA and tuberculosis.